Statistics Advance Part 1

Assignment Questions

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1. What is a random variable in probability theory?
2. What are the types of random variables?
3. What is the difference between discrete and continuous distributions?
4. What are probability distribution functions (PDF)?
5. How do cumulative distribution functions (CDF) differ from probability distribution functions (PDF)?
6. What is a discrete uniform distribution?
7. What are the key properties of a Bernoulli distribution?
8. What is the binomial distribution, and how is it used in probability?
9. What is the Poisson distribution and where is it applied?
10. What is a continuous uniform distribution?
11. What are the characteristics of a normal distribution?
12. What is the standard normal distribution, and why is it important?
13. What is the Central Limit Theorem (CLT), and why is it critical in statistics?
14. How does the Central Limit Theorem relate to the normal distribution?
15. What is the application of Z statistics in hypothesis testing?
16. How do you calculate a Z-score, and what does it represent?
17. What are point estimates and interval estimates in statistics?
18. What is the significance of confidence intervals in statistical analysis?
19. What is the relationship between a Z-score and a confidence interval?
20. How are Z-scores used to compare different distributions?
21. What are the assumptions for applying the Central Limit Theorem?
22. What is the concept of expected value in a probability distribution?
23. How does a probability distribution relate to the expected outcome of a random variable?